

Qibin Shi

Earthquake physics, soil hydrodynamics

(a) Education

Nanyang Technological University	Singapore	Geophysics	Ph.D., 2021 (December)
Nanjing University	Nanjing, China	Geology	B.Sc., 2016 (July)

(b) Professional Experience

2026 – present	Associate Professor	Institute of Geology and Geophysics, CAS
2024 – 2025	Pan Postdoctoral Fellow	Rice University
2022 – 2024	Postdoctoral Scholar	University of Washington

(c) Publications

Most closely related (* indicates the corresponding author)

1. **Qibin Shi***, David R Montgomery, Abigail LS Swann, Nicoleta C Cristea, Ethan F Williams, Nan You, Simon Jeffery, Joe Collins, Ana Prada Barrio, Paula A Misiewicz, *et al.*, Agroseismology and the impact of farming practices on soil hydrodynamics, *Science*, eaec0970 (2026).
2. Xinzhong Yin, **Qibin Shi***, Wenze Deng, Yingfeng Zhang, Zhangfeng Ma, Xiaoshu Li, Jiuhui Chen, Jinrong Su, and Shengji Wei, Concentrated rupture of the 2022 M_w 6.6 luding earthquake bounded by stress and geometry barriers, *Bulletin of the Seismological Society of America* (2025).
3. **Qibin Shi***, Ethan F Williams, Bradley P Lipovsky, Marine A Denolle, William SD Wilcock, Deborah S Kelley, and Katelyn Schoedl, Multiplexed distributed acoustic sensing offshore central oregon, *Seismological Research Letters* **96**, 784–800 (2025).
4. **Qibin Shi***, Marine A Denolle, Yiyu Ni, Ethan F Williams, and Nan You, Denoising offshore distributed acoustic sensing using masked auto-encoders to enhance earthquake detection, *Journal of Geophysical Research: Solid Earth* **130**, e2024JB029728 (2025).
5. **Qibin Shi*** and Marine A Denolle, Improved observations of deep earthquake ruptures using machine learning, *Journal of Geophysical Research: Solid Earth* **128**, e2023JB027334 (2023).
6. **Qibin Shi** and Shengji Wei*, Multiple point source inversion with regional broadband and local waveform data-a case study of the 2016 M_w 7.1 kumamoto, Japan earthquake, *Chinese Journal of Geophysics* **64**, 4378–4393 (2021).
7. **Qibin Shi*** and Shengji Wei, Highly heterogeneous pore fluid pressure enabled rupture of orthogonal faults during the 2019 ridgecrest M_w 7. 0 earthquake, *Geophysical Research Letters* **47**, e2020GL089827 (2020).
8. **Qibin Shi**, Sylvain Barbot*, Shengji Wei, Paul Tapponnier, Takanori Matsuzawa, and Bunichiro Shibazaki, Structural control and system-level behavior of the seismic cycle at the nankai trough, *Earth, Planets and Space* **72**, 27 (2020).
9. **Qibin Shi**, Shengji Wei*, and Meng Chen, An mcmc multiple point sources inversion scheme and its application to the 2016 kumamoto M_w 6.2 earthquake, *Geophysical Journal International* **215**, 737–752 (2018).
10. Teng Wang, **Qibin Shi**, Mehdi Nikkhoo, Shengji Wei, Sylvain Barbot, Douglas Dreger, Roland Bürgmann, Mahdi Motagh, and Qi-Fu Chen, The rise, collapse, and compaction of mt. mantap from the 3 september 2017 north korean nuclear test, *Science* **361**, 166–170 (2018).
11. Shengji Wei, Xin Wang, Chenglong Li, Hongyu Zeng, Zhangfeng Ma, **Qibin Shi**, Han Chen, Yihe Huang, Mingzhe Lyu, Jiangtao Liao, *et al.*, Supershear rupture sustained through a thick fault zone in the 2025 M_w 7.8 mandalay earthquake, *Science* **390**, 468–475 (2025).

Other significant publications

12. Marine A. Denolle*, **Qibin Shi**, Tim Clements, Loïc Viens, Veronica Rodriguez-Tribaldos, and Fabrice Cotton, Ambient field seismology in critical zone hydrological sciences, [Comptes Rendus Géoscience](#) **1**, 000–000 (2025), early view; Sciences de la Planète.
13. Yiyu Ni, Marine A Denolle, **Qibin Shi**, Bradley P Lipovsky, Shaowu Pan, and J Nathan Kutz, Wavefield reconstruction of distributed acoustic sensing: Lossy compression, wavefield separation, and edge computing, [JGR: Machine Learning and Computation](#) **1**, e2024JH000247 (2024).
14. Heng Luo, Hongyu Zeng, **Qibin Shi**, Teng Wang, Mingsheng Liao, Jiashun Hu, and Shengji Wei, Could thermal pressurization have induced the frequency-dependent rupture during the 2019 M_w 8.0 peru intermediate-depth earthquake?, [Geophysical Journal International](#) **232**, 115–127 (2023).
15. Shengji Wei, Hongyu Zeng, **Qibin Shi**, Jihong Liu, Heng Luo, Wanlin Hu, Yu Li, Weitao Wang, Zhangfeng Ma, Jing Liu-Zeng, *et al.*, Simultaneous rupture propagation through fault bifurcation of the 2021 M_w 7.4 maduo earthquake, [Geophysical Research Letters](#) **49**, e2022GL100283 (2022).
16. Irwan Meilano, Rino Salman, Suchi Rahmadani, **Qibin Shi**, Susilo Susilo, Eric Lindsey, Pepen Supendi, and Daryono Daryono, Source characteristics of the 2019 M_w 6.5 ambon, eastern indonesia, earthquake inferred from seismic and geodetic data, [Seismological Society of America](#) **92**, 3339–3348 (2021).

(d) Grants and Awards

2024 Pan Postdoc Fellowship, Earth, Environment, and Planetary Sciences, Rice University (\$7k)
 2019 Dr. Stephen Riady Geoscience Research Grant, Earth Observatory of Singapore (\$15k)

(e) Mentoring Experience

2023-present Mentor of undergrad student Hiroto Bito, University of Washington (paper in prep.)
 2023-present Mentor of student Xinzhong Yin, Earth Observatory of Singapore (published)

(f) Teaching Experience

2023 Guest Lecturer, *Plate Tectonics* (Marine Denolle), University of Washington
 2022 Guest Lecturer, *Machine Learning* (Marine Denolle), University of Washington
 2019 Teaching Assistant, *Scientific Writing* (Pavel Adamek), NTU
 2019 Teaching Assistant, *Natural Hazards & Society* (Kerry Sieh & Shengji Wei), NTU
 2019 Instructor, Seismological Workshop, Naypyidaw, Myanmar
 2018 Instructor, Seismological Workshop, Aceh, Indonesia
 2015 Instructor, Summer camp in an under-resourced community, Anhui, China

(g) Academic Service

Journal Reviews

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| • Earth and Space Science | • Seismica |
| • Earth, Planets and Space | • Seismological Research Letters |
| • Geophysical Journal International | • Journal of the Acoustical Society of America |
| • Geophysical Research Letters | • Bulletin of the Seismological Society of America |
| • Geoscience Letters | • IEEE Transactions on Geoscience and Remote Sensing |
| • Journal of Geophysical Research: Solid Earth | |
| • Science | |

Conferences

- Session convener at the American Geophysical Union Fall Meeting, 2025
- Session convener at the Seismological Society of America Annual Meeting, 2026